

120 Days of Quantitative Finance

Week 16: Stochastic Calculus + Time Series

Day 109: GARCH for Volatility Estimation

1 Introduction

GARCH models are widely used to estimate and forecast time-varying volatility in financial markets.

Unlike constant volatility models, GARCH captures clustering and persistence in volatility.

2 Model Setup

Let returns be:

$$r_t = \mu + \epsilon_t, \quad \epsilon_t = \sigma_t z_t$$

where:

$$z_t \sim \mathcal{N}(0, 1)$$

3 GARCH(1,1) Model

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

4 Interpretation

- α_1 : reaction to recent shocks
- β_1 : persistence of volatility
- $\alpha_1 + \beta_1$: overall persistence

5 Stationarity Condition

$$\alpha_1 + \beta_1 < 1$$

6 Volatility Forecasting

One-step ahead forecast:

$$\sigma_{t+1}^2 = \alpha_0 + \alpha_1 \epsilon_t^2 + \beta_1 \sigma_t^2$$

Multi-step forecast converges to:

$$\lim_{h \rightarrow \infty} \sigma_{t+h}^2 = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}$$

7 Volatility Clustering

- Large shocks increase future volatility
- Volatility decays slowly over time

8 Model Extensions

- EGARCH (asymmetric effects)
- GJR-GARCH (leverage effect)

9 Applications

- risk forecasting (VaR)

- portfolio optimization
- derivatives pricing

10 Conclusion

GARCH models provide a robust framework for estimating and forecasting volatility, capturing key empirical features of financial markets.